

MARATHON, FL, USA

WMO: 722016

Lat: 24.726N

Lon: 81.052W

Elev: 8

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 12896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.7	56.7	38.3	34.1	56.8	43.1	41.0	59.4	21.8	67.9	20.0	65.3	10.4	20	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.9	92.2	79.9	91.3	79.7	90.5	79.5	82.2	89.2	81.4	88.5	80.7	88.0	6.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	158.5	86.1	79.3	152.2	85.5	78.9	150.0	85.5	45.6	89.6	44.9	88.6	44.2	88.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.2	17.4	15.9	48.3	94.4	3.9	2.2	45.6	96.0	43.3	97.3	41.1	98.5	38.3	100.1		
			41.9	82.8	3.9	0.9	39.1	83.5	36.8	84.0	34.6	84.5	31.8	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.0	70.0	72.5	74.8	78.6	81.8	84.6	85.4	85.6	84.3	81.3	75.8	73.0
	DBStd	6.83	6.81	5.92	4.95	3.60	2.75	2.46	2.14	2.24	2.21	3.40	4.56	6.19
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	74	36	15	5	0	0	0	0	0	0	0	2	16
	CDD50	10589	621	631	769	857	985	1037	1098	1104	1029	971	774	713
	CDD65	5185	191	226	308	407	520	587	633	639	579	506	326	263
	CDH74	52556	764	1080	1824	3318	5516	7390	8280	8469	7225	5270	2060	1360
	CDH80	19173	55	122	275	755	1738	3215	3892	4091	3048	1568	285	129
Wind	WSAvg	8.0	8.8	8.2	8.8	8.8	8.0	6.6	6.2	6.4	6.9	9.0	9.3	8.8
Precipitation	PrecAvg	40.8	2.0	1.7	1.5	2.0	2.9	4.6	3.7	5.9	7.0	5.9	1.8	1.8
	PrecMax	55.4	8.2	6.1	3.5	6.1	9.1	14.7	8.3	19.3	15.2	17.2	6.7	5.4
	PrecMin	27.8	0.0	0.0	0.3	0.1	0.5	0.2	0.8	1.1	3.6	1.5	0.0	0.0
	PrecStd	7.4	2.0	1.6	1.0	1.5	2.4	3.3	1.9	3.3	2.4	4.1	1.8	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.5	85.3	86.4	89.3	91.0	93.3	93.3	93.2	92.0	89.9	87.0	84.8
		MCWB	74.3	75.0	76.4	77.8	78.4	79.7	80.1	80.4	80.2	79.6	77.5	76.3
	2%	DB	81.4	82.9	84.3	87.1	89.5	91.3	91.9	92.0	90.9	88.5	84.5	82.7
		MCWB	73.2	74.3	75.2	76.0	77.8	79.3	79.8	80.2	80.0	78.6	75.9	74.9
	5%	DB	80.1	81.4	82.9	85.6	88.1	90.4	91.0	91.2	90.1	87.7	82.9	81.4
		MCWB	72.3	73.2	74.2	75.0	76.9	79.1	79.6	80.0	79.7	78.1	74.9	74.0
	10%	DB	78.4	79.9	81.5	84.1	86.8	89.6	90.2	90.4	88.9	86.2	81.5	80.1
		MCWB	71.4	72.2	73.1	74.0	76.1	78.9	79.4	79.8	79.2	77.2	74.1	73.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.3	76.8	78.3	79.8	80.7	82.4	83.0	83.0	82.8	81.4	79.6	78.0
		MCDB	81.5	82.4	83.9	86.8	88.6	89.2	89.9	90.4	89.1	86.1	82.9	82.5
	2%	WB	75.0	75.4	76.7	77.9	79.2	81.3	81.7	82.0	81.5	80.3	78.2	76.5
		MCDB	79.7	81.3	82.5	84.5	87.1	88.5	89.1	89.6	88.2	85.5	82.4	80.8
	5%	WB	73.7	74.4	75.6	76.7	78.1	80.4	80.8	81.2	80.8	79.4	76.8	75.5
		MCDB	78.3	79.9	81.3	83.3	85.8	87.9	88.5	88.9	87.4	85.0	81.5	79.7
	10%	WB	72.6	73.3	74.5	75.6	77.3	79.7	80.1	80.5	80.1	78.4	75.6	74.5
		MCDB	77.1	78.5	80.2	82.3	84.8	87.5	88.0	88.1	86.7	84.1	80.3	78.9
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.0	10.5	9.9	9.5	9.5	9.7	9.9	9.5	8.5	8.6	8.9
		MCDBR	10.1	10.2	10.8	10.4	10.2	10.1	10.3	10.4	10.1	9.1	8.8	8.7
		MCWBR	5.7	5.5	5.8	5.3	4.8	4.6	4.6	4.9	4.7	4.4	4.6	5.2
	5% WB	MCDBR	10.3	9.9	10.7	10.0	9.6	9.9	9.9	10.2	9.6	8.7	8.8	8.3
		MCWBR	6.6	6.1	6.3	5.8	4.9	5.0	5.1	5.3	5.2	4.7	5.4	5.5
Clear-Sky Solar Irradiance	taub	0.363	0.378	0.384	0.402	0.425	0.445	0.452	0.435	0.424	0.409	0.379	0.371	
	taud	2.548	2.508	2.477	2.405	2.391	2.408	2.386	2.458	2.456	2.484	2.516	2.546	
	Ebn at Noon	284	287	290	285	276	269	267	272	274	273	276	277	
	Edh at Noon	29	32	35	38	39	38	39	36	35	33	30	28	
All-Sky Solar Radiation	RadAvg	1236	1525	1874	2114	2130	1976	2018	1919	1701	1528	1319	1164	
	RadStd	91	100	77	94	188	153	96	77	80	106	65	98	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.90	+1.93	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.68	+1.57	N/A	N/A	N/A	N/A	N/A	-23	+209	+187

Nomenclature: See separate page